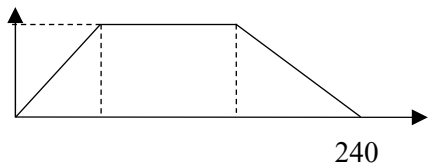
Question	Answer	Marks	Partial Marks
1(a)(i)	28	**2**	**M1** for $32 \times 0.50 + 30 \times 0.40$
1(a)(ii)	98 – 100 × 0.5 48 ÷ 0.4 = 120 [minutes] = 2 [hrs]	**M3**	**M1** for $100 \times 0.50 + x \times 0.40 = 98$ **M1** for $50 + 0.4x = 98$ or $0.4x = 48$ **M1** for $x = \frac{48}{0.4}$ $x = 120$ [min] = 2 [hr] OR **M1** for $100 \times 0.5 [= 50]$ **M1** for $98 - 50 [= 48]$ **M1** for $48 \div 0.4 = 120$ [min] = 2 [hr]
1(b)	2925 1170 4095	**3**	**B2** for one correct answer or **M1** for 8190 ÷ (5 + 2 + 7)
1(c)	58	**2**	**M1** for $\left(1 + \frac{45}{100}\right)k = 84.1$ oe
2(a)	0.18 or $\frac{9}{50}$	**1**	
2(b)	$1944 \times \frac{1000}{3600 \times 3600}$	**M1**	
	9 ÷ 0.15 = 60	**M1**	
2(c)		**1**	ruled line to axis with point of contact at 240
2(d)	6.9375	**4**	**M2** for area = $\frac{1}{2} \times (130 + 240) \times 9$ oe or **M1** for one correct partial area **M1dep** for *their* total area ÷ 240
3(a)	2.64 or 2.638…	**4**	**M3** for $[R^2 =] \frac{\pi \times 2.4^2 + \pi \times 2.4 \times 6.3}{\pi + 2\pi}$ oe or **M2** for $\pi \times 2.4^2 + \pi \times 2.4 \times 6.3 = \pi R^2 + \frac{1}{2} \times 4\pi R^2$ or **M1** for $[\pi \times 2.4^2] + \pi \times 2.4 \times 6.3$ oe or $[\pi R^2] + \frac{1}{2} \times 4\pi R^2$ oe

Question	Answer	Marks	Partial Marks
3(b)	953 or 952.6 to 952.8	**4**	**M3** for $\frac{1}{3}\times\pi\times7.6^2\times16\times\left(1-\left(\frac{16-12}{16}\right)^3\right)$ or $\frac{1}{3}\times\pi\times7.6^2\times16-\frac{1}{3}\times\pi\times1.9^2\times(16-12)$ OR **B1** for top radius = 1.9 or $\left(\frac{16-12}{16}\right)^3$ oe **M2** for $\frac{1}{3}\times\pi\times7.6^2\times16-\frac{1}{3}\times\pi\times(their\,1.9)^2\times(16-12)$ or $\frac{1}{3}\times\pi\times7.6^2\times16\times\left(1-their\left(\frac{16-12}{16}\right)^3\right)$ or **M1** for $\frac{1}{3}\times\pi\times7.6^2\times16$ or for $\frac{1}{3}\times\pi\times(their\,1.9)^2\times(16-12)$
4(a)(i)	438 cao	**2**	**M1** for $\frac{500}{1.142}$
4(a)(ii)	14.95	**2**	**M1** for [329 –] 275 × 1.142 oe
4(b)	14	**2**	**M1** for $5.25\times\frac{8}{3}$ oe
4(c)	1.7[0] or 1.699…	**3**	**M2** for $\sqrt[5]{\frac{6669}{6130}}$ or **M1** for 6669 = 6130 $(k)^5$
5(a)	13.5 or 13.47…	**4**	**B1** for angle 102 seen **M2** for $\sqrt{10.6^2+6.4^2-2\times10.6\times6.4\times\cos(180-78)}$ OR **M1** for $10.6^2+6.4^2-2\times10.6\times6.4\times\cos(180-78)$ **A1** for 181.5…
5(b)	8.68 or 8.682 to 8.683 nfww	**4**	**B1** for angle = 44 **M2** for sin(180 – 58 – 78) × $\frac{10.6}{\sin58}$ oe or **M1** for $\frac{\sin(180-58-78)}{x}=\frac{\sin58}{10.6}$ oe